



Personalized Valve Durability Prediction

A multimodal survival model for comparing candidate prosthetic valves

Patient/Valve • Physiology/Labs • Medications → Survival Curve

Objective: Given a patient's longitudinal history and a proposed valve, estimate model-based event-free durability.

Team Brute Force – Dyania 2nd Docathon

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Valve Durability in the Age of Personalized Medicine

From Medical Records to Operational Decisions

Clinical Need:

Prosthetic valve durability is a key consideration when choosing between feasible valve options.

Patients differ substantially in physiology, prior interventions, and treatment history.

The problem:

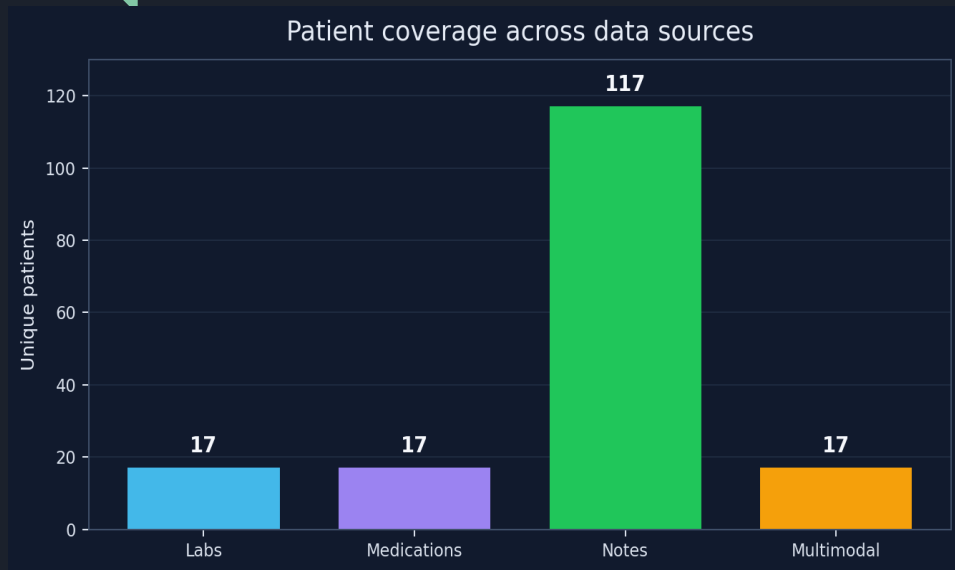
Durability estimates are largely population-level.

Clinicians still have to translate the evidence to the individual patient.

Our Solution:

A decision-support system that compares candidate valves for the same patient and estimates **personalized** event-free **durability** from longitudinal clinical data.

From fragmented clinical data to a model-ready cohort



117 note patients → 17 complete multimodal patients

Available Data

- 117 Patients with clinical Notes
- Longitudinal laboratory measurements
- Longitudinal medication records

Challenge

- Only 17 Patients with notes – labs – medication overlap.
- Irregular follow-up and missingness
- Aortic valve related notes mixed with free text in large medical notes

Our Approach

- Model patient lab and medication records as personal **longitudinal trajectories & observed valve characteristics**
- Use them to ground a semi-synthetic cohort for model development and validation.

From Real Clinical Data to Counterfactual Valve Outcomes

Grounding synthetic durability scenarios in real patient trajectories and observed valve configurations

Real Patient Foundation

- Laboratory and medical history
- Real missingness and follow-up patterns
- LLM – extracted prior valve history

Real candidate valves

- Observed SAVR and TAVR configurations
- Valve model, size and position
- **16 observed candidate configurations**

Synthetic outcome layer

- Patient-specific event-free durability
- Counterfactual outcomes for alternative valves
- Explicit **patient × valve interactions**

Real pre-operative patient history



Patient *i*
Longitudinal labs
Medications
Prior valve history
Real observation timing and missingness

Multiple candidate valves for the same patient

Valve A



Valve B



Valve C

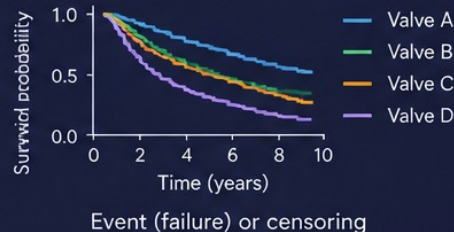


Valve D



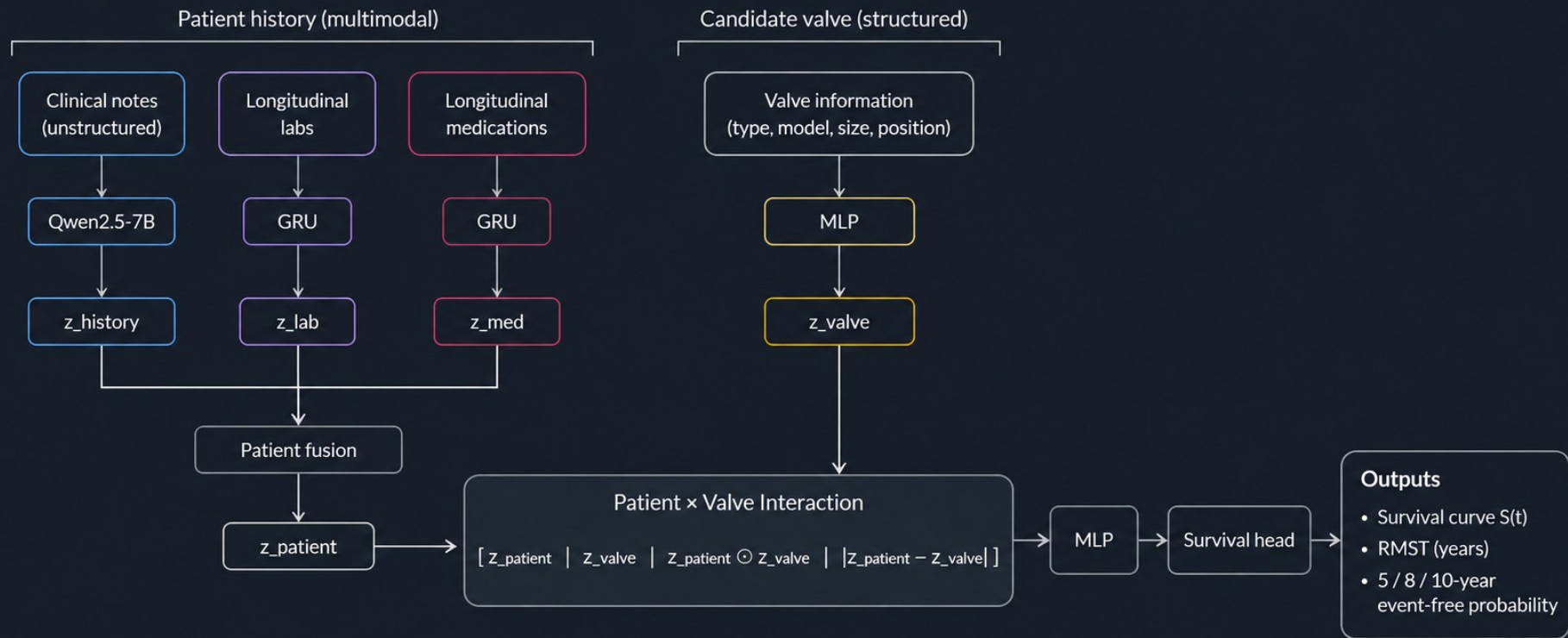
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Simulated post-implant outcomes



Interaction-aware multimodal survival model

Patient history and candidate valves are encoded separately then interacted to predict event - free survival





Validation of Personalized Valve Ranking

Held-out synthetic patient instances from the shared 17-patient real donor library

0.863

Harrell C-index
Overall survival
discrimination

0.823

Within-patient Spearman ρ
Ranking across candidate
valves

76.0%

Top-1 candidate recovery
Highest-durability
candidate identified

88.3%

Pairwise ranking accuracy
Correct ordering of two
candidate valves

Refocusing the objective on within-patient candidate ranking substantially improved personalized valve comparison.



Conclusions & Next Steps

What we demonstrated

- Built a multimodal pipeline from **clinical notes, longitudinal labs, medications, and valve characteristics**
- Modeled **patient × valve interactions** rather than only overall survival risk
- Achieved strong semi-synthetic ranking performance:
 - **88.3% pairwise ranking**
 - **76.0% top-1 candidate recovery**
- Produced interpretable outputs:
 - event-free durability in years
 - survival curves
 - patient drivers
 - valve-to-valve explanations

What remains

- Current outcomes are **semi-synthetic**
- Real donor library is small: **17 multimodal patients**
- No external clinical validation
- Future version needs:
 - larger real valve cohorts
 - richer anatomy/imaging and frailty data
 - competing-risk modeling
 - causal adjustment for treatment selection
 - external validation